### ✅ 1. Competency #7: Manipulate numbers, graphical and statistical data, algebraic expressions and equations, and geometric representations

- Highest relevance.
- The core of the task lies in precise numerical manipulation and comparison:
  - Correct sum: 155 + 86 + 79 = 320;
  - Difference: 320 − 248 = 72;
  - Simulating each error scenario numerically:
    - A (duplicate addition): e.g., 155 + 155 + 86 = 396 ≠ 248 → eliminated;
    - B (omitted item): 155+86=241, 155+79=234, 86+79=165 → none equal 248 → eliminated;
    - C (omitted last digit):
      - If 155 → 15: 15+86+79=180;
      - If 86 → 8: 155+8+79=242;
      - If 79 → 7: 155+86+7=248 → matches!
    - D (subtraction): 320 − 2×79 = 162, 320 − 2×86 = 148, 320 − 2×155 = 10 → none equal 248 → eliminated.
- This process involves systematic numerical transformation and verification, making Competency #7 the most directly engaged.

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### ✅ 2. Competency #12: Use a multi-step procedure leading to a mathematical solution, conclusion or generalisation

- Highly relevant.
- Solving requires a clear sequence of steps:
  1. Compute the correct total;
  2. Note the incorrect result;
  3. Analyze the magnitude of the discrepancy;
  4. Model each hypothesis and compute its outcome;
  5. Compare results and identify the unique match.
- This constitutes a structured diagnostic procedure, where each step depends on the previous one—fully aligning with Competency #12.

### ✅ 3. Competency #13: Use an understanding of the context to guide or expedite the mathematical solving process

- Highly relevant.
- The context— "calculator input error" —guides students toward plausible human mistakes:
  - "Omitting the last digit" corresponds to missing a key press (e.g., typing "7" instead of "79");
  - "Adding twice" reflects accidentally pressing '+' or re-entering a number;
  - "Subtracting" suggests hitting '−' by mistake.
- This real-world framing helps students focus on realistic errors and avoid implausible assumptions (e.g., decimal point errors). Contextual understanding narrows the solution space and improves efficiency.

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### ✅ 4. Competency #8: Articulate a solution, showing and/or summarising and presenting intermediate mathematical results

- Implicitly expected.
- Although the task is multiple-choice, in educational or assessment settings, students should be able to explain why option C is correct, for example:

> "If Olivia entered 79 as 7 (omitting the last digit), then 155 + 86 + 7 = 248, which exactly matches her result."

- This causal justification demonstrates deep understanding of the error mechanism and aligns with Competency #8.

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### ✅ 5. Competency #1: Perform a simple calculation

- Foundational support.
- Involves multi-digit addition and subtraction (e.g., 155+86+79, 320−248). While basic, accuracy is critical.
- A single arithmetic error (e.g., miscalculating the total as 310) would derail the entire reasoning chain. Thus, reliable basic computation is essential.

### ✅ 1. 能力 #7：操作数字、图表与统计信息、代数式、方程以及几何表示

- 最高相关性。
- 本题核心在于对数字进行精确操作与比较：
  - 正确求和：155 + 86 + 79 = 320；
  - 计算差值：320 − 248 = 72；
  - 对每个选项进行数值模拟：
    - A（重复加）：例如 155+155+86=396 ≠ 248 → 排除；
    - B（漏加）：155+86=241，155+79=234，86+79=165 → 均 ≠ 248 → 排除；
    - C（漏最后一位）：如将 155 误作 15，则 15+86+79=180；若 86→8，则 155+8+79=242；若 79→7，则 155+86+7=248 → 匹配！
    - D（减去一项）：320 − 2×79 = 162，320 − 2×86 = 148，320 − 2×155 = 10 → 均 ≠ 248 → 排除。
- 整个过程涉及系统性的数值变换与验证，是能力 #7 的典型体现。

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### ✅ 2. 能力 #12：使用多步骤程序得出数学解、结论或一般化结果

- 高度相关。
- 解题需清晰的多阶段流程：
  1. 计算正确总和；
  2. 观察错误结果；
  3. 分析偏差大小；
  4. 对四个假设逐一建模并计算；
  5. 比较结果，锁定唯一匹配项。
- 这是一个结构化的诊断程序，每一步都依赖前一步，完全符合能力 #12。

### ✅ 3. 能力 #13：利用对情境的理解来指导或加速数学解题过程

- 高度相关。
- 情境关键词是“计算器输入错误”，这引导学生关注常见人为操作失误：
  - “漏掉最后一位”对应按键遗漏（如输入 79 时只按了 7）；
  - “加了两次”对应重复按“+”或数字；
  - “减去”对应误触“−”键。
- 这种现实背景帮助学生聚焦合理错误类型，避免不切实际的猜测（如小数点错误）。情境理解直接缩小了验证范围并提升了效率。

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### ✅ 4. 能力 #8：清晰表述解题过程，展示和/或总结并呈现中间数学结果

- 隐含要求。
- 虽然题目是选择题，但在教学或评估中，学生应能解释为何选 C，例如：

  > “如果 Olivia 把 79 误输为 7（漏掉最后一位），那么 155 + 86 + 7 = 248，正好匹配她的结果。”

- 这种因果论证体现了对错误机制的理解，符合能力 #8 的精神。

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### ✅ 5. 能力 #1：执行简单计算

- 基础支撑。
- 涉及多位数加减法（155+86+79, 320−248 等），虽为基础技能，但准确性决定成败。
- 若计算出错（如误算总和为 310），会导致整个推理链崩溃。因此，可靠的基本运算能力是前提。